

QuNorth Operator Azure and Quantum OS setup guide

This guide helps Magne Operators at QuNorth set up the Azure tenant, subscriptions, and resources needed to connect to and use the Magne device. The guide has three phases.

Phase	Description	Target date
1	Set up the Azure tenant, subscriptions, resource groups, and resource types needed to use Magne	September 2026
2	Connect to Magne, configure and verify logs and alerts, and Quantum OS training	November 2026
3	Ongoing administrative responsibilities and device monitoring	Ongoing

This guide provides the steps for creating and configuring the Azure resources used to operate Magne and Quantum OS. It also identifies optional recommendations based on Azure best practices. QuNorth is responsible for architectural and configuration decisions beyond the requirements explicitly stated in this guide.

If you have any questions while going through these steps, contact your Microsoft Quantum Support team by email: quantum_support@microsoft.com

To get started with Azure architecture and security fundamentals, see the following Azure documentation.

- [Azure Well-Architected Framework](#)
- [Azure application architecture fundamentals](#)
- [Azure security best practices and patterns](#)

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Phase 1: Initial Azure tenant setup and Azure resource creation for Magne

In the first phase, you configure your Azure tenant, set security rules, assign roles to users, and create subscriptions and resource groups. Then, you create and configure the two Azure resource types that are required to operate Magne:

- Provider Account
- Log Analytics workspace

Part 1: Validate your Microsoft Entra ID tenant

QuNorth already operates an Entra ID tenant through a managed Microsoft 365 service. In this part, you verify the existing tenant configurations and apply a small set of baseline identity settings before you create Azure resources.

Part 3: Create a Provider Account and Configure Log Analytics

In this part, QuNorth creates a Provider Account in Azure, verifies that the account is created successfully, completes the final configuration with the Microsoft Quantum team, configures diagnostic logging, and sets up monitoring.

What is a Provider Account?

An Azure Quantum Provider Account is a special type of Azure resource that represents a quantum computing device - such as an atom-based quantum processor - that is integrated with the Azure Quantum platform.

After you set up a Provider Account, researchers can connect to it through an Azure Quantum workspace and submit quantum jobs.

A Provider Account can:

- Surface one or more physical quantum device targets and emulator or simulator targets.
- Expose multiple target modes, such as physical-qubit and logical-qubit modes.
- Control which Azure subscriptions can access the device.
- Define quota allocation per subscription.
- Be managed by an Operator who is responsible for the hardware.

Prerequisites

Before you begin, you need:

- The QuNorth Azure environment set up for Azure Quantum.
- An active Azure subscription, and the subscription ID for that subscription.
- The **Contributor** role on the Azure subscription or the target resource group.
- Azure CLI installed and signed in to the correct Azure account and subscription. For installation instructions, see [How to install the Azure CLI](#).
- An Azure subscription that's allowlisted for the **Provider Account** resource type. To request allowlisting, provide your Azure subscription ID to the Microsoft Azure Quantum team and ask them to enable the **Provider Account** resource type for your subscription.

Step 1: Create an Azure Resource Manager (ARM) template

1. Open a plain-text editor, such as Visual Studio Code or Windows Notepad, and create a new file. Copy the example ARM template in this section into the file, update the required values for your environment, and save it as `provider-account-template.json`. Don't use Microsoft Word or another rich-text editor, because it can add formatting that makes the JSON invalid.
2. Create a dedicated local folder for the deployment files, such as `C:\Magne\ARM`, and save the template in that folder as `provider-account-template.json`. Confirm that the file name ends in `.json`, not `.json.txt`.

ARM template properties

Property	Example value	Description
<code>type</code>	<code>microsoft.quantum/provideraccounts</code>	The Azure resource type to create. In this case, it's a Provider Account under the <code>microsoft.quantum</code> resource provider.
<code>apiVersion</code>	<code>2025-02-15-preview</code>	The version of the Azure Resource Manager API used to manage this resource. Use the version provided by the Microsoft Quantum team.
<code>name</code>	<code>pa-qunorth-swedencentral-001</code>	A unique, human-readable name for the Provider Account. Use a descriptive name, such as the device location and number.
<code>location</code>	<code>swedencentral</code>	The Azure region closest to the physical location of the quantum device.
<code>identity.type</code>	<code>SystemAssigned</code>	Configures Azure to automatically create a managed identity for the resource. This identity authenticates backend services securely without storing credentials.
<code>providerId</code>	<code>qunorth</code>	A unique identifier for the quantum provider. The length must be between 2 and 14 characters. Only lowercase letters, dashes, and numbers are allowed. This ID appears in job submissions and must be unique across all Azure Quantum regions.
<code>displayName</code>	<code>QuNorth</code>	The friendly name shown to researchers in the Quantum OS portal and workspace.
<code>description</code>	<code>QuNorth - A Nordic quantum initiative</code>	A short description of the device that is shown to researchers when they browse available providers.
<code>companyName</code>	<code>QuNorth</code>	The name of the company or institution that owns and operates the quantum device.

Example ARM template

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "resources": [
```

```

{
  "type": "microsoft.quantum/provideraccounts",
  "apiVersion": "2025-02-15-preview",
  "name": "pa-qunorth-swedencentral-001",
  "location": "swedencentral",
  "identity": {
    "type": "SystemAssigned"
  },
  "properties": {
    "providerId": "qunorth",
    "displayName": "QuNorth",
    "description": "QuNorth - A Nordic quantum initiative",
    "companyName": "QuNorth"
  }
}
]
}

```

Important fields to customize for your deployment

Make sure that you update the following fields in the ARM template for your environment:

- `name`: Use a name that reflects your device and location.
- `location`: Use the Azure region closest to your device, such as `swedencentral`.
- `providerId`: Use a short identifier that's unique across all Azure Quantum regions (up to 14 alphanumeric characters).
- `displayName` and `description`: Use clear, descriptive text that researchers see.

Step 2: Deploy the ARM template

Use Azure CLI to deploy the ARM template to a resource group.

Before running these commands, open a terminal and go to the folder where you saved `provider-account-template.json`. Keep the same terminal session open for the remaining Azure CLI steps.

```

cd C:\Magne\ARM
dir provider-account-template.json

```

Before running these commands, replace the placeholder values in angle brackets (< >) with your own values.

1. Sign in to Azure, select the Magne subscription, and verify that the terminal session is using the correct tenant and subscription. These commands are also used earlier in this guide when creating the resource group.

```
az login
```

```
az account set --subscription "<qunorth-magne-prod-subscription-id>"
```

```
az account show --output table
```

Complete the browser sign-in and multifactor authentication prompt if requested. In the verification output, confirm that the subscription is `qunorth-magne-prod` and that the tenant ID matches the QuNorth Entra tenant before you continue.

2. If the resource group doesn't exist, create it.

```
az group create \
  --name <rg-magne-provider-logs> \
  --location <swedencentral>
```

3. Deploy the ARM template to your resource group.

```
az deployment group create \
  --resource-group <rg-magne-provider-logs> \
  --template-file <provider-account-template.json>
```

Wait for the deployment to complete. When the deployment finishes, Azure CLI returns a JSON response confirming the result. Look for `"provisioningState": "Succeeded"` in the output.

When the Provider Account is created successfully, Azure Quantum completes several setup steps automatically. No additional action is required for these steps:

- Azure Quantum registers the provider and its targets so allowlisted subscriptions can discover them in Quantum OS.
- Researchers in allowlisted subscriptions can see the device in Quantum OS.

Step 3: Complete Provider Account configuration

After verifying the successful creation of the Provider Account, work with the Microsoft Quantum team to complete the Provider Account configuration.

Work with the Microsoft Quantum team to configure the following items:

- The quantum device and its internal metadata.
- The quantum device targets.
- Access to Quantum OS so that you can validate the configured devices, queues, and targets.

IMPORTANT

Access to Quantum OS isn't available until a future date. You can't perform some setup validation steps until you have access to Quantum OS.

Step 4: Create a Log Analytics workspace

After you complete the Provider Account configuration, create a Log Analytics workspace to store Provider Account logs.

You can create the Log Analytics workspace in the Azure portal or by using the Azure CLI.

The Log Analytics workspace stores Provider Account logs for troubleshooting, review, and basic operational monitoring.

NOTE

The Log Analytics workspace incurs storage costs from Azure. This cost is the only cost that QuNorth incurs for the Azure resources in this guide. If you decide to use Azure resources other than the ones in this guide, you might incur additional costs for those services.

Recommended settings

Use the following settings when you create the workspace:

- Create the workspace in the same Azure region and subscription as the Provider Account.
- Dedicate the workspace to only Quantum OS logs for the Provider Account. Don't use the workspace for unrelated Azure resources.
- Use a clear name, for example, `law-magne-provider-swedencentral-001`.
- Keep the default retention setting unless your team requires a longer retention period.
- Keep public network access enabled. This article assumes that the workspace receives logs through the standard Azure public endpoint.
- Grant access only to people who need to view or troubleshoot Provider Account logs, such as the Operator team.

Azure portal

1. Sign in to the Azure portal with an account that has permission to create resources in the target resource group.
2. In the Azure portal, search for and select **Log Analytics workspaces**.
3. Select **Create**.
4. Select the subscription (same as Provider Account) and resource group where you want to create the workspace.
5. Enter a workspace name, for example, `law-magne-provider-swedencentral-001`.
6. Select the same Azure region as the Provider Account.
7. Select **Review + create**, and then select **Create**.

Azure CLI

Run the following commands from the same Azure CLI session that you used to deploy and verify the Provider Account. Before you run the commands, replace the placeholder values in angle brackets (`<>`) with your own values.

```
az account set --subscription "<provider-subscription-id>"
```

```
az monitor log-analytics workspace create \
  --resource-group rg-magne-provider-logs \
  --workspace-name law-magne-provider-swedencentral-001 \
  --location swedencentral --sku PerGB2018
```

Step 5: Configure Provider Account diagnostic settings

To retain and query Magne logs, configure diagnostic settings on the Provider Account resource and send the selected log categories to the Log Analytics workspace that you created earlier.

You can configure Provider Account diagnostic settings in the Azure portal or by using the Azure CLI.

Azure portal

1. Sign in to the Azure portal.
2. Open the Provider Account resource that you created earlier.
3. In the left navigation pane, select **Diagnostic settings**.
4. Select **Add diagnostic setting**.
5. Enter a diagnostic setting name, such as `ds-provideraccount-to-law`.
6. Select the Provider Account log categories that are relevant for your operations, auditing, and troubleshooting needs.
7. Under **Destination details**, select **Send to Log Analytics workspace**.
8. Select the subscription and Log Analytics workspace that you created earlier.
9. Select **Save**.

Azure CLI

Use the following commands to configure diagnostic settings for the Provider Account. Replace the placeholder values in angle brackets (< >) with your own values.

```
PROVIDER_ACCOUNT_ID="/subscriptions/<subscription-id>/resourceGroups/rg-magne-provider-logs/providers/microsoft.quantum/provideraccounts/pa-qunorth-swedencentral-001"
```

```
WORKSPACE_ID="/subscriptions/<subscription-id>/resourceGroups/rg-magne-provider-logs/providers/Microsoft.OperationalInsights/workspaces/law-magne-provider-swedencentral-001"
```

```
az monitor diagnostic-settings categories list --resource "$PROVIDER_ACCOUNT_ID"
```

```
az monitor diagnostic-settings create `
  --name ds-provideraccount-to-law `
  --resource "$PROVIDER_ACCOUNT_ID" `
  --workspace "$WORKSPACE_ID" `
  --logs '[{"categoryGroup\":\"allLogs\", \"enabled\":true}]'
```

NOTE

Validation of your Log Analytics configurations and setup of alerts for Magne occurs in Phase 2, after access to the Quantum OS is available.

Step 6: Apply resource locks

Protect the Provider Account and Log Analytics resources from accidental deletion.

IMPORTANT

To manage locks, you need `Microsoft.Authorization/locks/*` permissions. The **Owner** and **User Access Administrator** roles have these permissions, but the **Contributor** role doesn't.

Apply CanNotDelete lock

1. In the Azure portal, open the `rg-magne-provider-logs` resource group, select **Locks**, and then select **Add**.
2. Configure the lock: enter `prevent-deletion` for **Lock name**, select **Delete (CanNotDelete)** for **Lock type**, and add notes as appropriate.

NOTE

Use the **CanNotDelete** lock for this resource group. Avoid **ReadOnly** locks on actively managed resources - **ReadOnly** blocks routine operations (for example, regenerating keys or updating diagnostic settings), not just deletion.

Verify lock

1. Open a locked resource group, select **Locks**, and confirm that the `prevent-deletion` lock is present.
2. Attempt to delete a resource in a locked group and confirm that the action is blocked.
3. Document which resource groups have locks.

NOTE

If you delete and recreate a resource group, you remove locks. Subscription cancellation bypasses locks. If QuNorth adopts infrastructure as code, define the locks in Bicep or Terraform so they're versioned and reapplied.

Related documentation

- [Create a Log Analytics workspace in the Azure portal](#)
- [Overview of Log Analytics in Azure Monitor](#)
- [Lock resources to protect infrastructure](#)